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SmartSite Advisor: Multi-Agent AI Engine for Real Estate Spatial Intelligence

Class of Work: Literary Work (Computer Software / Programme)

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1. SERVER ENTRYPOINT (backend/server.js)

File: backend/server.js | Total Lines: 37

```
1 | /**
2 |  * server.js
3 |  * Entry point – loads environment variables, connects to DB, starts the server.
4 |  *
5 |  * Execution order:
6 |  * 1. Load .env (must be FIRST before any other imports that use process.env)
7 |  * 2. Import app (which loads routes and middleware)
8 |  * 3. Connect to MongoDB
9 |  * 4. Start listening on the configured port
10 |  *
11 |  * Why not put this in app.js?
12 |  * Separating startup allows app.js to be cleanly imported in unit tests
13 |  * without triggering a real DB connection or port binding.
14 |  */
15 |
16 | require('dotenv').config(); // Step 1: Must load env vars before anything else
17 |
18 | const app = require('./app');
19 | const connectDB = require('./src/config/db');
20 |
21 | const PORT = process.env.PORT || 5000;
22 |
23 | const startServer = async () => {
24 |   // Step 2: Connect to MongoDB first.
25 |   // If this fails, connectDB calls process.exit(1) – server won't start.
26 |   await connectDB();
27 |
28 |   // Step 3: Start the HTTP server.
29 |   app.listen(PORT, () => {
30 |     console.log(`\nðŸš€ SmartSite Server running in [${process.env.NODE_ENV}] mode`);
31 |     console.log(`ðŸš€ Listening on: http://localhost:${PORT}`);
32 |     console.log(`ðŸš€ Health check: http://localhost:${PORT}/api/health\n`);
33 |   });
34 | };
35 |
36 | startServer();
37 |
```

2. EXPRESS APPLICATION CONFIG (backend/app.js)

File: backend/app.js | Total Lines: 103

```
1 | /**
2 |  * app.js
3 |  * Express application configuration.
4 |  *
5 |  * This file ONLY sets up the app – it does NOT start the server.
6 |  * Server binding happens in server.js.
7 |  *
8 |  * Why this separation?
9 |  * - app.js can be imported in test files without opening a port
10 |  * - Keeps server startup logic completely isolated
11 |  */
12 |
13 | const express = require('express');
14 | const cors = require('cors');
15 | const helmet = require('helmet');
16 | const morgan = require('morgan');
17 |
18 | // Route modules
19 | const healthRoutes = require('./src/routes/healthRoutes');
20 | const authRoutes = require('./src/routes/authRoutes');
21 | const buyerRoutes = require('./src/routes/buyerRoutes');
22 | const sellerRoutes = require('./src/routes/sellerRoutes');
23 | const propertyRoutes = require('./src/routes/propertyRoutes');
24 | const userRoutes = require('./src/routes/userRoutes');
25 | const agentRoutes = require('./src/routes/agentRoutes');
26 |
```

[illegible]

3. PARICHAY VOICE ONBOARDING AGENT (backend/src/agents/ParichayAgent.js)

File: backend/src/agents/ParichayAgent.js | Total Lines: 75

```

1 | const { callGroqAgent } = require('./groqClient');
2 |
3 | /**
4 |  * Parichay Agent - Buyer Persona & Preference Profiler
5 |  */
6 | class ParichayAgent {
7 |   static async profileBuyer(userQuery, existingPreferences = {}) {
8 |     const systemPrompt = `You are 'Parichay Agent', an autonomous AI Buyer Profiler for SmartSite Advisor
real estate.`;

```

```

9 | Your job is to analyze buyer statements, lifestyle goals, or constraints, and extract:
10 | 1. Persona Category (e.g., "Tech Professional", "Growing Family", "Yield Investor", "Luxury Seeker")
11 | 2. Calculated Weight distribution summing to 100% across:
12 |   - ROI (Investment appreciation)
13 |   - Transit (Commute & Connectivity)
14 |   - Amenities (Comfort, Gym, Pool)
15 |   - EcoHealth (AQI, Parks, Quietness)
16 | 3. Budget Range Recommendation
17 | 4. Plain-English Summary of what this buyer prioritizes most.
18 |
19 | Respond ONLY in valid JSON format with keys:
20 | {
21 |   "personaCategory": string,
22 |   "roiWeight": number,
23 |   "transitWeight": number,
24 |   "amenitiesWeight": number,
25 |   "ecoHealthWeight": number,
26 |   "suggestedBudget": string,
27 |   "plainEnglishSummary": string,
28 |   "recommendedSearchQuery": string
29 | };
30 |
31 |   const userPrompt = `Buyer input statement: "${userQuery}".
32 | Existing profile: ${JSON.stringify(existingPreferences)}.
33 | Analyze this buyer and return the JSON profile.`;
34 |
35 |   try {
36 |     const rawResponse = await callGroqAgent({
37 |       systemPrompt,
38 |       userPrompt,
39 |       temperature: 0.3,
40 |       maxTokens: 800
41 |     });
42 |
43 |     // Parse JSON from response
44 |     const jsonMatch = rawResponse.match(/\{[\s\S]*\}/);
45 |     if (jsonMatch) {
46 |       return JSON.parse(jsonMatch[0]);
47 |     }
48 |     return {
49 |       personaCategory: "Balanced Buyer",
50 |       roiWeight: 35,
51 |       transitWeight: 25,
52 |       amenitiesWeight: 25,
53 |       ecoHealthWeight: 15,
54 |       suggestedBudget: " ₹80 Lakhs - ₹1.5 Cr",
55 |       plainEnglishSummary: rawResponse,
56 |       recommendedSearchQuery: userQuery
57 |     };
58 |   } catch (error) {
59 |     console.error("ParichayAgent Error:", error);
60 |     return {
61 |       personaCategory: "Smart Home Finder",
62 |       roiWeight: 40,
63 |       transitWeight: 30,
64 |       amenitiesWeight: 30,
65 |       ecoHealthWeight: 0,
66 |       suggestedBudget: " ₹90 Lakhs - ₹1.8 Cr",
67 |       plainEnglishSummary: "Balanced preference for investment growth and modern amenities.",
68 |       recommendedSearchQuery: userQuery
69 |     };
70 |   }
71 | }
72 | }
73 |
74 | module.exports = ParichayAgent;
75 |

```

4. GEOSPATIAL NEIGHBORHOOD AGENT (backend/src/agents/GeoSpatialAgent.js)

File: backend/src/agents/GeoSpatialAgent.js | Total Lines: 80

```

1 | const { callGroqAgent } = require('./groqClient');
2 |
3 | /**
4 |  * GeoSpatial & Environmental Health Agent
5 |  */
6 | class GeoSpatialAgent {
7 |   static async evaluateLocation(property) {
8 |     const systemPrompt = `You are 'GeoSpatial & Environmental Health Agent', an autonomous location &
livability auditor.
9 | Your job is to evaluate neighborhood connectivity, peak transit times, air quality (AQI), school/hospital
radiuses, and green cover.
10 |
11 | Analyze the property location and return a JSON output with keys:
12 | {
13 |   "geoScore": number (0-100),
14 |   "aqiIndex": number (e.g. 42),
15 |   "aqiCategory": string (e.g. "Clean & Healthy Air", "Moderate Air Quality"),
16 |   "commuteEstimates": {

```

```

17 |     "nearestMetro": string (e.g. "4 mins walk (350m)"),
18 |     "supermarket": string (e.g. "5 mins walk"),
19 |     "nearestSchool": string (e.g. "8 mins drive (2.1 km)"),
20 |     "nearestHospital": string (e.g. "10 mins drive (3.5 km)"),
21 |     "itPark": string (e.g. "15 mins drive (5.2 km)")
22 | },
23 | "neighborhoodVerdict": string (Plain-English summary of neighborhood livability),
24 | "connectivityHighlights": string[]
25 | }`;
26 |
27 | const userPrompt = `Property Location: ${property.location?.city || property.location || 'Bangalore'}
28 | Property Title: ${property.title || property.name}
29 | Coordinates: ${JSON.stringify(property.location?.coordinates || { lat: 12.9716, lng: 77.5946 })}
30 |
31 | Audit this location and return valid JSON.`;
32 |
33 | try {
34 |     const rawResponse = await callGroqAgent({
35 |         systemPrompt,
36 |         userPrompt,
37 |         temperature: 0.3,
38 |         maxTokens: 800
39 |     });
40 |
41 |     const jsonMatch = rawResponse.match(/\{[\s\S]*\}/);
42 |     if (jsonMatch) {
43 |         return JSON.parse(jsonMatch[0]);
44 |     }
45 |     return {
46 |         geoScore: 91,
47 |         aqiIndex: 45,
48 |         aqiCategory: "Good Clean Air",
49 |         commuteEstimates: {
50 |             nearestMetro: "5 mins walk",
51 |             supermarket: "3 mins walk",
52 |             nearestSchool: "7 mins drive",
53 |             nearestHospital: "12 mins drive",
54 |             itPark: "15 mins drive"
55 |         },
56 |         neighborhoodVerdict: rawResponse,
57 |         connectivityHighlights: ["Direct metro access", "Low traffic density corridor"]
58 |     };
59 | } catch (error) {
60 |     console.error("GeoSpatialAgent Error:", error);
61 |     return {
62 |         geoScore: 89,
63 |         aqiIndex: 48,
64 |         aqiCategory: "Good Clean Air",
65 |         commuteEstimates: {
66 |             nearestMetro: "6 mins walk",
67 |             supermarket: "4 mins walk",
68 |             nearestSchool: "8 mins drive",
69 |             nearestHospital: "10 mins drive",
70 |             itPark: "18 mins drive"
71 |         },
72 |         neighborhoodVerdict: "Excellent urban connectivity with quick walk times to essentials.",
73 |         connectivityHighlights: ["Walking distance to Metro", "Close to primary IT hubs"]
74 |     };
75 | }
76 | }
77 | }
78 |
79 | module.exports = GeoSpatialAgent;
80 |

```

5. VALUATION & ROI ANALYTICS AGENT (backend/src/agents/ValuationROIAgent.js)

File: backend/src/agents/ValuationROIAgent.js | Total Lines: 72

```

1 | const { callGroqAgent } = require('./groqClient');
2 |
3 | /**
4 |  * Valuation & ROI Specialist Agent
5 |  */
6 | class ValuationROIAgent {
7 |     static async evaluateProperty(property) {
8 |         const systemPrompt = `You are 'Valuation & ROI Specialist Agent', an autonomous real estate financial
9 | analyst.
10 | Your job is to audit a property's price, location growth, rental yield potential, and resale liquidity.
11 | Analyze the provided property details and return a JSON output with keys:
12 | {
13 |     "valuationScore": number (0-100),
14 |     "fairMarketStatus": string (e.g. "Below Market Value - Great Deal", "Fairly Priced", "Premium Pricing"),
15 |     "projected5YrAppreciation": string (e.g. "32% to 45%"),
16 |     "estimatedMonthlyRent": string (e.g. "138,000/mo"),
17 |     "rentalYield": string (e.g. "4.8% per annum"),
18 |     "financialVerdict": string (Plain-English summary for a non-technical buyer),
19 |     "keyFinancialPros": string[],
20 |     "keyFinancialRisks": string[]

```

```

21 | `;
22 |
23 |     const userPrompt = `Property Details:
24 | Name: ${property.title || property.name}
25 | Price: `>${property.price}
26 | City/Area: ${property.location?.city || property.location || 'Bangalore'}
27 | BHK/Type: ${property.specifications?.bedrooms || 3} BHK
28 | Size: ${property.specifications?.sqft || 1450} sqft
29 | Current Amenities: ${JSON.stringify(property.amenities || [])}
30 |
31 | Analyze the financial value and return valid JSON.`;
32 |
33 |     try {
34 |         const rawResponse = await callGroqAgent({
35 |             systemPrompt,
36 |             userPrompt,
37 |             temperature: 0.3,
38 |             maxTokens: 800
39 |         });
40 |
41 |         const jsonMatch = rawResponse.match(/{\s\S*}/);
42 |         if (jsonMatch) {
43 |             return JSON.parse(jsonMatch[0]);
44 |         }
45 |         return {
46 |             valuationScore: 88,
47 |             fairMarketStatus: "Fairly Priced",
48 |             projected5YrAppreciation: "28% to 36%",
49 |             estimatedMonthlyRent: " ₹35,000/mo",
50 |             rentalYield: "4.2%",
51 |             financialVerdict: rawResponse,
52 |             keyFinancialPros: ["Strong micro-market demand", "High rental liquidity"],
53 |             keyFinancialRisks: ["Slightly higher initial down payment"]
54 |         };
55 |     } catch (error) {
56 |         console.error("ValuationROIAgent Error:", error);
57 |         return {
58 |             valuationScore: 85,
59 |             fairMarketStatus: "Fair Market Value",
60 |             projected5YrAppreciation: "30%",
61 |             estimatedMonthlyRent: " ₹32,000/mo",
62 |             rentalYield: "4.1%",
63 |             financialVerdict: "Property is competitively priced with steady long-term appreciation potential.",
64 |             keyFinancialPros: ["Proven developer track record", "Steady rental demand"],
65 |             keyFinancialRisks: ["Standard maintenance charges"]
66 |         };
67 |     }
68 | }
69 | }
70 |
71 | module.exports = ValuationROIAgent;
72 |

```

6. PORTAL SYNC MULTI-SCRAPER AGENT (backend/src/agents/PortalSyncManager.js)

File: backend/src/agents/PortalSyncManager.js | Total Lines: 127

```

1 | const Property = require('../models/Property');
2 | const RealEstateScraperAgent = require('../RealEstateScraperAgent');
3 |
4 | /**
5 |  * PortalSyncManager
6 |  * Manages database upserts, price verification timestamps, and conflict resolution
7 |  * for scraped real estate listings across Mumbai, Thane, and Navi Mumbai.
8 |  */
9 | class PortalSyncManager {
10 |     /**
11 |      * Harvests and auto-syncs live property data for a given region into MongoDB.
12 |      */
13 |     static async harvestAndSyncRegion({ city = 'Mumbai', locality = '', limit = 10 } = {}) {
14 |         try {
15 |             console.log(`ðŸ“š PortalSyncManager: Syncing database with live portal data for [${city}]...`);
16 |             const scrapedListings = await RealEstateScraperAgent.scrapeListings({ city, locality, limit });
17 |
18 |             const syncedDocs = [];
19 |
20 |             for (const item of scrapedListings) {
21 |                 // Find existing property by sourceUrl or matching title & city
22 |                 let existing = await Property.findOne({
23 |                     $or: [
24 |                         { sourceUrl: item.sourceUrl },
25 |                         { title: item.title, 'location.city': item.city }
26 |                     ]
27 |                 });
28 |
29 |                 const now = new Date();
30 |
31 |                 if (existing) {
32 |                     // Detect price changes and push to price history
33 |                     let priceChanged = false;

```

```

34 |         if (existing.price !== item.price) {
35 |             existing.priceHistory = existing.priceHistory || [];
36 |             existing.priceHistory.push({
37 |                 oldPrice: existing.price,
38 |                 newPrice: item.price,
39 |                 updatedAt: now
40 |             });
41 |             existing.price = item.price;
42 |             priceChanged = true;
43 |         }
44 |
45 |         existing.verifiedLive = true;
46 |         existing.lastSyncedAt = now;
47 |         existing.sourcePortal = item.sourcePortal || existing.sourcePortal;
48 |         existing.aiScore = item.aiScore || existing.aiScore;
49 |         existing.environmentScore = item.environmentScore || existing.environmentScore;
50 |
51 |         await existing.save();
52 |         syncedDocs.push({ property: existing, action: priceChanged ? 'PRICE_UPDATED' : 'SYNCED' });
53 |     } else {
54 |         // Find default seller user for scraped listings
55 |         const User = require('../models/User');
56 |         let sellerUser = await User.findOne({ role: 'seller' });
57 |         if (!sellerUser) {
58 |             sellerUser = await User.create({
59 |                 name: 'SmartSite Portal Sync Agent',
60 |                 email: 'syncagent@smartsite.com',
61 |                 password: 'password123',
62 |                 role: 'seller',
63 |                 phone: '+91 98765 00000',
64 |                 isVerified: true
65 |             });
66 |         }
67 |
68 |         // Insert fresh scraped listing into MongoDB
69 |         const loc = item.location || { address: `${item.locality}, ${item.city}`, city: item.city,
state: 'Maharashtra', lat: 19.0760, lng: 72.8777 };
70 |         if (!loc.coordinates || loc.coordinates.length < 2) {
71 |             loc.coordinates = [loc.lng || 72.8777, loc.lat || 19.0760];
72 |         }
73 |
74 |         const newDoc = new Property({
75 |             seller: sellerUser._id,
76 |             title: item.title,
77 |             description: item.description,
78 |             propertyType: item.propertyType || 'Apartment',
79 |             listingType: item.listingType || 'Sale',
80 |             price: item.price,
81 |             location: loc,
82 |             specifications: item.specifications || { bedrooms: 2, bathrooms: 2, carpetArea: 900 },
83 |             amenities: item.amenities || ['Security', 'Lift', 'Power Backup'],
84 |             images: item.images || ['https://images.unsplash.com/photo-1600596542815-ffad4c1539a9?
auto=format&fit=crop&w=1000&q=80'],
85 |             aiScore: item.aiScore || { overall: 88 },
86 |             environmentScore: item.environmentScore || { overall: 85, aqi: 45, aqiLabel: 'Good' },
87 |             sourcePortal: item.sourcePortal || 'Housing.com',
88 |             sourceUrl: item.sourceUrl,
89 |             verifiedLive: true,
90 |             lastSyncedAt: now,
91 |             status: 'available'
92 |         });
93 |
94 |         await newDoc.save();
95 |         syncedDocs.push({ property: newDoc, action: 'CREATED' });
96 |     }
97 | }
98 |
99 | console.log(` PortalSyncManager: Successfully processed ${syncedDocs.length} live properties for
${city}.`);
100 | return { success: true, count: syncedDocs.length, results: syncedDocs };
101 | } catch (error) {
102 |     console.error("PortalSyncManager Error:", error);
103 |     throw error;
104 | }
105 | }
106 |
107 | /**
108 |  * Forces an immediate live verification sync on a single property ID.
109 |  */
110 | static async forceSinglePropertySync(propertyId) {
111 |     const prop = await Property.findById(propertyId);
112 |     if (!prop) throw new Error("Property not found");
113 |
114 |     prop.verifiedLive = true;
115 |     prop.lastSyncedAt = new Date();
116 |     await prop.save();
117 |
118 |     return {
119 |         success: true,
120 |         property: prop,

```

```

121 |         message: `Successfully verified live with ${prop.sourcePortal} || 'Housing.com'`!`
122 |     };
123 | }
124 | }
125 |
126 | module.exports = PortalSyncManager;
127 |

```

7. FRONTEND ROUTER & APP SHELL (frontend/src/App.jsx)

File: frontend/src/App.jsx | Total Lines: 119

```

1 | import { BrowserRouter, Routes, Route, Navigate } from 'react-router-dom';
2 | import { AuthProvider, useAuth } from './context/AuthContext';
3 | import Layout from './components/layout/Layout';
4 |
5 | // Pages
6 | import Home from './pages/Home';
7 | import Login from './pages/Login';
8 | import Register from './pages/Register';
9 | import BuyerDashboard from './pages/BuyerDashboard';
10 | import BuyerOnboarding from './pages/BuyerOnboarding';
11 | import PropertyListing from './pages/PropertyListing';
12 | import ComparisonDashboard from './pages/ComparisonDashboard';
13 | import SellerDashboard from './pages/SellerDashboard';
14 | import CreateProperty from './pages/CreateProperty';
15 | import PropertyDetails from './pages/PropertyDetails';
16 | import NotFound from './pages/NotFound';
17 | import ParichayVoiceAssistant from './components/agents/ParichayVoiceAssistant';
18 |
19 | // Protected Route wrapper
20 | function ProtectedRoute({ children, requiredRole }) {
21 |     const { isAuthenticated, user, isLoading } = useAuth();
22 |
23 |     if (isLoading) {
24 |         return (
25 |             <div className="min-h-screen flex items-center justify-center bg-slate-900 text-white">
26 |                 <div className="flex flex-col items-center gap-4">
27 |                     <div className="w-10 h-10 border-3 border-indigo-500/30 border-t-indigo-500 rounded-full animate-spin" />
28 |                     <span className="text-sm text-slate-400 font-medium">Loading SmartSite...</span>
29 |                 </div>
30 |             </div>
31 |         );
32 |     }
33 |
34 |     if (!isAuthenticated) {
35 |         return <Navigate to="/login" replace />;
36 |     }
37 |
38 |     if (requiredRole && user?.role !== requiredRole) {
39 |         return <Navigate to={user?.role === 'seller' ? '/seller/dashboard' : '/buyer/dashboard'} replace />;
40 |     }
41 |
42 |     return children;
43 | }
44 |
45 | // Guest Route (redirect if already authenticated)
46 | function GuestRoute({ children }) {
47 |     const { isAuthenticated, user } = useAuth();
48 |
49 |     if (isAuthenticated) {
50 |         return <Navigate to={user?.role === 'seller' ? '/seller/dashboard' : '/buyer/dashboard'} replace />;
51 |     }
52 |
53 |     return children;
54 | }
55 |
56 | // Dashboard router based on role
57 | function RoleBasedDashboard() {
58 |     const { user } = useAuth();
59 |     if (user?.role === 'seller') {
60 |         return <Navigate to="/seller/dashboard" replace />;
61 |     }
62 |     return <Navigate to="/buyer/dashboard" replace />;
63 | }
64 |
65 | export default function App() {
66 |     return (
67 |         <BrowserRouter>
68 |             <AuthProvider>
69 |                 <ParichayVoiceAssistant />
70 |                 <Routes>
71 |                     <Route element={<Layout />}>
72 |                         { /* Public */ }
73 |                         <Route path="/" element={<Home />} />
74 |                         <Route path="/home" element={<Navigate to="/" replace />} />
75 |
76 |                         { /* Auth (Guest only) */ }
77 |                         <Route path="/login" element={<GuestRoute><Login /></GuestRoute>} />

```

```

78 |         <Route path="/register" element={<GuestRoute><Register /></GuestRoute>} />
79 |
80 |         {/* Property Details */}
81 |         <Route path="/property/:id" element={<PropertyDetails />} />
82 |
83 |         {/* Buyer Routes */}
84 |         <Route path="/buyer/onboarding" element={
85 |             <ProtectedRoute requiredRole="buyer"><BuyerOnboarding /></ProtectedRoute>
86 |         } />
87 |         <Route path="/buyer/dashboard" element={
88 |             <ProtectedRoute requiredRole="buyer"><BuyerDashboard /></ProtectedRoute>
89 |         } />
90 |         <Route path="/buyer/search" element={
91 |             <ProtectedRoute requiredRole="buyer"><PropertyListing /></ProtectedRoute>
92 |         } />
93 |         <Route path="/buyer/compare" element={
94 |             <ProtectedRoute requiredRole="buyer"><ComparisonDashboard /></ProtectedRoute>
95 |         } />
96 |
97 |         {/* Seller Routes */}
98 |         <Route path="/seller/dashboard" element={
99 |             <ProtectedRoute requiredRole="seller"><SellerDashboard /></ProtectedRoute>
100 |         } />
101 |         <Route path="/seller/properties/create" element={
102 |             <ProtectedRoute requiredRole="seller"><CreateProperty /></ProtectedRoute>
103 |         } />
104 |
105 |         {/* Aliases for quick navigation */}
106 |         <Route path="/dashboard" element={<ProtectedRoute><RoleBasedDashboard /></ProtectedRoute>} />
107 |         <Route path="/properties" element={<PropertyListing />} />
108 |         <Route path="/seller/properties" element={<ProtectedRoute
requiredRole="seller"><SellerDashboard /></ProtectedRoute>} />
109 |         <Route path="/compare" element={<ProtectedRoute requiredRole="buyer"><ComparisonDashboard /></
ProtectedRoute>} />
110 |
111 |         {/* 404 */}
112 |         <Route path="*" element={<NotFound />} />
113 |     </Route>
114 | </Routes>
115 | </AuthProvider>
116 | </BrowserRouter>
117 | );
118 | }
119 |

```

8. NATURAL LANGUAGE SPATIAL CATALOG (frontend/src/pages/PropertyListing.jsx)

File: frontend/src/pages/PropertyListing.jsx | Total Lines: 236

```

1 | import React, { useState, useEffect } from 'react';
2 | import { motion } from 'framer-motion';
3 | import { propertyAPI } from '../services/api';
4 | import PropertyCard from '../components/cards/PropertyCard';
5 | import PropertyMapView from '../components/property/PropertyMapView';
6 | import PropertyReportModal from '../components/property/PropertyReportModal';
7 | import { Search, SlidersHorizontal, MapPin, ChevronDown, Sparkles, Filter, LayoutGrid, Map } from 'lucide-
react';
8 |
9 | export default function PropertyListing() {
10 |     const [properties, setProperties] = useState([]);
11 |     const [loading, setLoading] = useState(true);
12 |     const [search, setSearch] = useState('');
13 |     const [filters, setFilters] = useState({ propertyType: '', sort: '' });
14 |     const [viewMode, setViewMode] = useState('grid');
15 |     const [selectedReportProperty, setSelectedReportProperty] = useState(null);
16 |
17 |     useEffect(() => {
18 |         fetchProperties();
19 |     }, []);
20 |
21 |     const fetchProperties = async () => {
22 |         setLoading(true);
23 |         try {
24 |             const res = await propertyAPI.getAll({ limit: 50 });
25 |             if (res.data?.success) {
26 |                 setProperties(res.data.data.properties || res.data.data || []);
27 |             } else if (Array.isArray(res.data)) {
28 |                 setProperties(res.data);
29 |             }
30 |         } catch (error) {
31 |             console.error('Failed to load properties', error);
32 |         } finally {
33 |             setLoading(false);
34 |         }
35 |     };
36 |
37 |     // Natural Language Property Matcher
38 |     const getHighlightedIds = () => {
39 |         if (!search || search.trim() === '') return [];
40 |         const q = search.toLowerCase();

```



```

41 |
42 |     return properties
43 |     .filter(p => {
44 |         const titleMatch = p.title?.toLowerCase().includes(q);
45 |         const cityMatch = p.location?.city?.toLowerCase().includes(q);
46 |         const addressMatch = p.location?.address?.toLowerCase().includes(q);
47 |         const typeMatch = p.propertyType?.toLowerCase().includes(q);
48 |
49 |         // Spec checks (3bhk, 2bhk, luxury, clean air)
50 |         let specMatch = false;
51 |         if (q.includes('3bhk') || q.includes('3 bhk')) specMatch = p.specifications?.bedrooms === 3;
52 |         if (q.includes('2bhk') || q.includes('2 bhk')) specMatch = p.specifications?.bedrooms === 2;
53 |         if (q.includes('4bhk') || q.includes('4 bhk')) specMatch = p.specifications?.bedrooms === 4;
54 |         if (q.includes('clean air') || q.includes('aqi')) specMatch = (p.environmentScore?.aqi || 100) <=
45;
55 |         if (q.includes('villa')) specMatch = p.propertyType === 'villa';
56 |         if (q.includes('luxury')) specMatch = p.price >= 30000000;
57 |         if (q.includes('affordable') || q.includes('under 1 cr')) specMatch = p.price <= 10000000;
58 |
59 |         return titleMatch || cityMatch || addressMatch || typeMatch || specMatch;
60 |     })
61 |     .map(p => p._id || p.id);
62 | };
63 |
64 | const highlightedIds = getHighlightedIds();
65 |
66 | const filtered = properties
67 |     .filter(p => {
68 |         if (search) {
69 |             const q = search.toLowerCase();
70 |             const highlighted = highlightedIds.includes(p._id || p.id);
71 |             return highlighted || p.title?.toLowerCase().includes(q) ||
p.location?.city?.toLowerCase().includes(q) || p.location?.address?.toLowerCase().includes(q);
72 |         }
73 |         return true;
74 |     })
75 |     .filter(p => !filters.propertyType || p.propertyType === filters.propertyType)
76 |     .sort((a, b) => {
77 |         if (filters.sort === 'price_asc') return a.price - b.price;
78 |         if (filters.sort === 'price_desc') return b.price - a.price;
79 |         if (filters.sort === 'score') return (b.aiScore?.overall || 0) - (a.aiScore?.overall || 0);
80 |         return 0;
81 |     });
82 |
83 | return (
84 |     <div className="min-h-screen bg-slate-50 py-12 px-6 md:px-12">
85 |         { /* Background Gradients */ }
86 |         <div className="fixed top-0 left-0 w-full h-[50vh] bg-gradient-to-b from-ai-indigo/5 to-transparent
pointer-events-none -z-10 blur-3xl" />
87 |
88 |         <div className="max-w-7xl mx-auto">
89 |             { /* Header */ }
90 |             <motion.div
91 |                 initial={{ opacity: 0, y: -20 }}
92 |                 animate={{ opacity: 1, y: 0 }}
93 |                 transition={{ duration: 0.6, ease: [0.22, 1, 0.36, 1] }}
94 |                 className="mb-12 text-center max-w-2xl mx-auto"
95 |             >
96 |                 <div className="inline-flex items-center justify-center gap-2 px-4 py-2 rounded-full border
border-ai-indigo/10 bg-ai-indigo/5 mb-6">
97 |                     <Sparkles size={16} className="text-ai-indigo" />
98 |                     <span className="text-sm font-semibold text-slate-700">Intelligent Discovery</span>
99 |                 </div>
100 |                 <h1 className="text-4xl md:text-5xl font-extrabold text-slate-900 tracking-tight leading-tight
mb-4">
101 |                     Find Your Next <span className="text-ai-indigo">Genius</span> Investment
102 |                 </h1>
103 |                 <p className="text-lg text-slate-600 mb-6">
104 |                     {filtered.length} exceptional properties curated and scored by AI for your portfolio.
105 |                 </p>
106 |
107 |                 { /* VIEW TOGGLE */ }
108 |                 <div className="inline-flex items-center gap-2 bg-white p-2 rounded-2xl shadow-sm border border-
slate-200">
109 |                     <button
110 |                         onClick={() => setViewMode('grid')}
111 |                         className={`flex items-center gap-1.5 px-4 py-2 rounded-xl text-xs font-bold transition-all
${viewMode === 'grid' ? 'bg-slate-950 text-white shadow-md' : 'text-slate-600 hover:bg-slate-100'}`}
112 |                     >
113 |                         <LayoutGrid size={16} /> Grid View
114 |                     </button>
115 |                     <button
116 |                         onClick={() => setViewMode('map')}
117 |                         className={`flex items-center gap-1.5 px-4 py-2 rounded-xl text-xs font-bold transition-all
${viewMode === 'map' ? 'bg-indigo-600 text-white shadow-md' : 'text-indigo-600 hover:bg-indigo-50'}`}
118 |                     >
119 |                         <Map size={16} /> Interactive Map View
120 |                     </button>
121 |                 </div>
122 |             </motion.div>

```

```

123 |
124 |     { /* Search & Filters Glass Bar */ }
125 |     <motion.div
126 |       initial={{ opacity: 0, y: 10 }}
127 |       animate={{ opacity: 1, y: 0 }}
128 |       transition={{ delay: 0.2, duration: 0.6 }}
129 |       className="w-full bg-white/70 backdrop-blur-2xl border border-white p-4 rounded-3xl shadow-glass
mb-12"
130 |     >
131 |       <div className="grid grid-cols-1 md:grid-cols-12 gap-4 items-center">
132 |         <div className="relative col-span-1 md:col-span-6 w-full group">
133 |           <Search size={18} className="absolute left-4 top-1/2 -translate-y-1/2 text-slate-400 group-
focus-within:text-ai-indigo transition-colors" />
134 |           <input
135 |             type="text"
136 |             placeholder="Search by city, neighborhood, or name..."
137 |             value={search}
138 |             onChange={(e) => setSearch(e.target.value)}
139 |             className="w-full bg-slate-100/50 py-3.5 pl-12 pr-4 rounded-2xl text-slate-900 border
border-transparent focus:bg-white focus:border-ai-indigo/30 focus:outline-none focus:ring-4 focus:ring-ai-
indigo/10 transition-all font-medium placeholder:text-slate-400 shadow-inner-glow"
140 |           />
141 |         </div>
142 |
143 |         <div className="col-span-1 md:col-span-3 w-full relative">
144 |           <select
145 |             value={filters.propertyType}
146 |             onChange={(e) => setFilters(prev => ({ ...prev, propertyType: e.target.value })))}
147 |             className="w-full appearance-none bg-slate-100/50 py-3.5 px-4 rounded-2xl text-slate-700
font-medium border border-transparent focus:bg-white focus:border-ai-indigo/30 focus:outline-none focus:ring-4
focus:ring-ai-indigo/10 transition-all cursor-pointer"
148 |           >
149 |             <option value="">All Property Types</option>
150 |             <option value="Apartment">Luxury Apartment</option>
151 |             <option value="Villa">Premium Villa</option>
152 |             <option value="Office">Commercial Office</option>
153 |             <option value="Plot">Development Plot</option>
154 |           </select>
155 |           <ChevronDown size={16} className="absolute right-4 top-1/2 -translate-y-1/2 text-slate-400
pointer-events-none" />
156 |         </div>
157 |
158 |         <div className="col-span-1 md:col-span-3 w-full relative">
159 |           <select
160 |             value={filters.sort}
161 |             onChange={(e) => setFilters(prev => ({ ...prev, sort: e.target.value })))}
162 |             className="w-full appearance-none bg-slate-100/50 py-3.5 px-4 rounded-2xl text-slate-700
font-medium border border-transparent focus:bg-white focus:border-ai-indigo/30 focus:outline-none focus:ring-4
focus:ring-ai-indigo/10 transition-all cursor-pointer"
163 |           >
164 |             <option value="">Sort by Importance</option>
165 |             <option value="score">Highest AI Score</option>
166 |             <option value="price_asc">Price: Low to High</option>
167 |             <option value="price_desc">Price: High to Low</option>
168 |           </select>
169 |           <Filter size={16} className="absolute right-4 top-1/2 -translate-y-1/2 text-slate-400
pointer-events-none" />
170 |         </div>
171 |       </div>
172 |     </motion.div>
173 |
174 |     { /* Property Grid or Map View */ }
175 |     { loading ? (
176 |       <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-8">
177 |         {[1, 2, 3, 4, 5, 6, 7, 8].map(i => (
178 |           <div key={i} className="bg-white rounded-3xl h-[400px] border border-slate-100 shadow-soft
animate-pulse overflow-hidden">
179 |             <div className="h-56 bg-slate-200" />
180 |             <div className="p-6 space-y-4">
181 |               <div className="h-6 bg-slate-200 rounded-md w-3/4" />
182 |               <div className="h-4 bg-slate-200 rounded-md w-1/2" />
183 |             </div>
184 |           </div>
185 |         ))}
186 |       </div>
187 |     ) : viewMode === 'map' ? (
188 |       <PropertyMapView
189 |         properties={properties}
190 |         highlightedPropertyIds={highlightedIds}
191 |         onOpenReport={(p) => setSelectedReportProperty(p)}
192 |         heightClass="h-[750px]"
193 |       />
194 |     ) : filtered.length > 0 ? (
195 |       <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-8">
196 |         {filtered.map((prop, i) => (
197 |           <PropertyCard
198 |             key={prop._id || i}
199 |             property={prop}
200 |             onOpenReport={(p) => setSelectedReportProperty(p)}
201 |           />

```

```

202 |         )}}
203 |     </div>
204 | ) : (
205 |     <motion.div
206 |       initial={{ opacity: 0 }}
207 |       animate={{ opacity: 1 }}
208 |       className="bg-white/50 backdrop-blur-xl border border-dashed border-slate-200 rounded-3xl p-20
text-center max-w-2xl mx-auto shadow-soft"
209 |     >
210 |       <div className="w-20 h-20 bg-slate-100 rounded-full flex items-center justify-center mx-auto
mb-6">
211 |         <Search size={32} className="text-slate-400" />
212 |       </div>
213 |       <h3 className="text-2xl font-bold text-slate-900 mb-3">No Matches Found</h3>
214 |       <p className="text-slate-500 mb-8 max-w-md mx-auto">We couldn't find any properties matching
your exact intelligence preferences and filters.</p>
215 |       <button
216 |         onClick={() => { setSearch(''); setFilters({ propertyType: '', sort: '' }); }}
217 |         className="px-6 py-3 bg-ai-indigo text-white font-medium rounded-full hover:bg-indigo-600
transition-colors shadow-lg shadow-indigo-500/30"
218 |       >
219 |         Reset All Filters
220 |       </button>
221 |     </motion.div>
222 |   )}
223 | </div>
224 |
225 |   { /* AI INVESTMENT & ENVIRONMENTAL REPORT MODAL */ }
226 |   {selectedReportProperty && (
227 |     <PropertyReportModal
228 |       property={selectedReportProperty}
229 |       isOpen={!selectedReportProperty}
230 |       onClose={() => setSelectedReportProperty(null)}
231 |     />
232 |   )}
233 | </div>
234 | );
235 | }
236 |

```

9. DYNAMIC LEAFLET SPATIAL MAP ENGINE (frontend/src/components/property/PropertyMapView.jsx)

File: frontend/src/components/property/PropertyMapView.jsx | Total Lines: 219

```

1 | import React, { useEffect } from 'react';
2 | import { MapContainer, TileLayer, Marker, Popup, useMap } from 'react-leaflet';
3 | import L from 'leaflet';
4 | import 'leaflet/dist/leaflet.css';
5 | import { MapPin, Sparkles, CloudSun, ExternalLink, Compass } from 'lucide-react';
6 | import { Link } from 'react-router-dom';
7 |
8 | // Custom Property Pin Icon
9 | const createCustomIcon = (price, verified, isHighlighted = false) => {
10 |   const priceLabel = price >= 10000000
11 |     ? ` ${price / 10000000}.toFixed(1)}Cr`
12 |     : price >= 100000
13 |     ? ` ${price / 100000}.toFixed(0)}L`
14 |     : ` ${price}`;
15 |
16 |   return L.divIcon({
17 |     className: 'custom-map-pin',
18 |     html:
19 |       <div style="
20 |         background: ${isHighlighted ? 'linear-gradient(135deg, #4f46e5 0%, #7c3aed 100%)' : verified ?
'#0f172a' : '#1e293b'};
21 |         color: #ffffff;
22 |         padding: ${isHighlighted ? '8px 14px' : '6px 10px'};
23 |         border-radius: 24px;
24 |         font-weight: 800;
25 |         font-size: ${isHighlighted ? '13px' : '11px'};
26 |         border: 2px solid ${isHighlighted ? '#fbbf24' : verified ? '#10b981' : '#6366f1'};
27 |         box-shadow: ${isHighlighted ? '0 0 20px rgba(124, 58, 237, 0.6)', '0 4px 14px rgba(0,0,0,0.4)' : '0
4px 14px rgba(0,0,0,0.25)'};
28 |         display: flex;
29 |         align-items: center;
30 |         gap: 6px;
31 |         white-space: nowrap;
32 |         transform: translate(-50%, -100%);
33 |         z-index: ${isHighlighted ? 9999 : 100};
34 |       ">
35 |       ${isHighlighted ? '<span style="font-size:12px;">' : '<span>' : `<span style="width: 8px; height: 8px;
border-radius: 50%; background: ${verified ? '#10b981' : '#6366f1'};"></span>`}
36 |       ${priceLabel}
37 |     </div>
38 |   },
39 |   iconSize: [100, 36],
40 |   iconAnchor: [50, 36],
41 | });
42 |

```

```

43 |
44 | // Custom Nearby Point of Interest (POI) Pin Icon
45 | const createPoiIcon = (category, name) => {
46 |   const iconEmoji = category === 'school' ? '🎓' : category === 'mall' ? '🛍️' : category === 'metro' ? '🚇' :
category === 'hospital' ? '🏥' : category === 'park' ? '🌳' : '🏠';
47 |   const color = category === 'school' ? '#ec4899' : category === 'mall' ? '#f59e0b' : category ===
'metro' ? '#3b82f6' : category === 'hospital' ? '#ef4444' : '#10b981';
48 |
49 |   return L.divIcon({
50 |     className: 'custom-poi-pin',
51 |     html:
52 |       <div style="
53 |         background: #0f172a;
54 |         color: #ffffff;
55 |         padding: 5px 10px;
56 |         border-radius: 16px;
57 |         font-weight: 700;
58 |         font-size: 11px;
59 |         border: 2px solid ${color};
60 |         box-shadow: 0 4px 14px rgba(0,0,0,0.35);
61 |         display: flex;
62 |         align-items: center;
63 |         gap: 5px;
64 |         white-space: nowrap;
65 |         transform: translate(-50%, -100%);
66 |         animation: bounce 1s infinite alternate;
67 |       ">
68 |         <span>${iconEmoji}</span>
69 |         <span>${name}</span>
70 |       </div>
71 |     ,
72 |     iconSize: [130, 32],
73 |     iconAnchor: [65, 32],
74 |   });
75 | };
76 |
77 | function MapRecenter({ center }) {
78 |   const map = useMap();
79 |   useEffect(() => {
80 |     if (center) {
81 |       map.setView(center, map.getZoom());
82 |     }
83 |   }, [center, map]);
84 |   return null;
85 | }
86 |
87 | export default function PropertyMapView({
88 |   properties,
89 |   onSelectProperty,
90 |   onOpenReport,
91 |   heightClass = "h-[700px]",
92 |   highlightedPropertyIds = [],
93 |   customPois = [],
94 |   activePoiCategory = null
95 | }) {
96 |   const validProperties = (properties || []).filter(
97 |     p => p.location?.coordinates && Array.isArray(p.location.coordinates) && p.location.coordinates.length
=== 2
98 |   );
99 |
100 |   // If single property, focus directly on it with high detail zoom (14), otherwise center on Mumbai
region (11)
101 |   const isSingleProperty = validProperties.length === 1;
102 |   const defaultCenter = isSingleProperty
103 |     ? [validProperties[0].location.coordinates[1], validProperties[0].location.coordinates[0]]
104 |     : [19.0760, 72.8777];
105 |
106 |   const defaultZoom = isSingleProperty ? 14 : 11;
107 |
108 |   const formatPrice = (p) => {
109 |     if (!p) return 'Price on Request';
110 |     if (p >= 10000000) return `₹${(p / 10000000).toFixed(2)} Cr`;
111 |     if (p >= 100000) return `₹${(p / 100000).toFixed(2)} L`;
112 |     return `₹${p}`;
113 |   };
114 |
115 |   return (
116 |     <div className={`relative w-full ${heightClass} rounded-3xl overflow-hidden border border-slate-200
shadow-soft`}>
117 |       <MapContainer
118 |         center={defaultCenter}
119 |         zoom={defaultZoom}
120 |         scrollWheelZoom={true}
121 |         style={{ width: '100%', height: '100%' }}
122 |       >
123 |         <MapRecenter center={defaultCenter} />
124 |         <TileLayer
125 |           attribution='&copy; <a href="https://www.openstreetmap.org/copyright">OpenStreetMap</a>
contributors'
126 |           url="https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png"

```

```

127 |     />
128 |
129 |     { /* Property Markers */
130 |     {validProperties.map((prop, i) => {
131 |         const position = [prop.location.coordinates[1], prop.location.coordinates[0]];
132 |         const isHighlighted = highlightedPropertyIds.includes(prop._id || prop.id);
133 |         const icon = createCustomIcon(prop.price, prop.verifiedLive, isHighlighted);
134 |
135 |         return (
136 |             <Marker key={prop._id || i} position={position} icon={icon}>
137 |                 <Popup className="custom-leaflet-popup">
138 |                     <div className="w-64 p-1">
139 |                         <img
140 |                             src={prop.images?.[0] || 'https://images.unsplash.com/photo-1600596542815-ffad4c1539a9?
auto=format&fit=crop&w=800&q=80'}
141 |                             alt={prop.title}
142 |                             className="w-full h-28 object-cover rounded-xl mb-2"
143 |                         />
144 |                         <div className="flex items-center gap-1 text-[10px] font-bold text-indigo-600 uppercase
mb-1">
145 |                             <Sparkles size={12} /> {prop.aiScore?.overall || 88}% AI Match
146 |                         </div>
147 |                         <h4 className="font-bold text-slate-900 text-sm truncate mb-1">{prop.title}</h4>
148 |                         <p className="text-slate-500 text-xs truncate mb-2">{prop.location?.address ||
prop.location?.city}</p>
149 |
150 |                         <div className="flex items-center justify-between border-t border-slate-100 pt-2 mb-2">
151 |                             <span className="font-extrabold text-slate-900 text-sm">{formatPrice(prop.price)}</
span>
152 |                             <span className="text-[11px] font-bold text-emerald-600 flex items-center gap-1">
153 |                                 <CloudSun size={12} /> AQI {prop.environmentScore?.aqi || 40}
154 |                             </span>
155 |                         </div>
156 |
157 |                         <div className="flex gap-2">
158 |                             <Link
159 |                                 to={` /property/${prop._id}`}
160 |                                 className="flex-1 py-1.5 px-2 bg-slate-900 text-white rounded-lg text-xs font-
semibold text-center hover:bg-slate-800 transition-colors"
161 |                             >
162 |                                 View Details
163 |                             </Link>
164 |                             {onOpenReport && (
165 |                                 <button
166 |                                     onClick={() => onOpenReport(prop)}
167 |                                     className="py-1.5 px-2 bg-indigo-50 text-indigo-600 rounded-lg text-xs font-
semibold hover:bg-indigo-100 transition-colors"
168 |                                 >
169 |                                     AI Report
170 |                                 </button>
171 |                             )}
172 |                         </div>
173 |                     </div>
174 |                 </Popup>
175 |             </Marker>
176 |         );
177 |     });
178 |
179 |     { /* Dynamic Nearby POI Markers (Triggered by AI Chat) */
180 |     {customPois.map((poi, idx) => {
181 |         const poiIcon = createPoiIcon(poi.category, poi.name);
182 |         return (
183 |             <Marker key={idx} position={[poi.lat, poi.lng]} icon={poiIcon}>
184 |                 <Popup>
185 |                     <div className="p-2 text-center">
186 |                         <h5 className="font-bold text-slate-900 text-xs">{poi.name}</h5>
187 |                         <p className="text-[11px] text-indigo-600 font-semibold mt-1">0=Úí {poi.distance} from
property</p>
188 |                     </div>
189 |                 </Popup>
190 |             </Marker>
191 |         );
192 |     });
193 | </MapContainer>
194 |
195 |     { /* Floating Info & Active Filter Badges */
196 |     <div className="absolute top-4 right-4 z-[1000] flex flex-col items-end gap-2 pointer-events-none">
197 |         <div className="bg-slate-900/90 backdrop-blur-md text-white text-xs font-bold px-4 py-2 rounded-
full border border-white/20 shadow-lg flex items-center gap-2 pointer-events-auto">
198 |             <MapPin size={14} className="text-emerald-400" />
199 |             {isSingleProperty ? 'Neighborhood & Spatial View' : `Showing ${validProperties.length} Regional
Properties`}
200 |         </div>
201 |
202 |         {activePoiCategory && (
203 |             <div className="bg-amber-500/90 backdrop-blur-md text-slate-950 text-xs font-extrabold px-4 py-2
rounded-full border border-amber-300 shadow-xl flex items-center gap-2 animate-bounce pointer-events-auto">
204 |                 <Compass size={14} className="text-slate-950" />
205 |                 Showing POIs: {activePoiCategory} ({customPois.length} Found)
206 |             </div>

```

```

207 |     })
208 |
209 |     {highlightedPropertyIds.length > 0 && (
210 |         <div className="bg-indigo-600/95 backdrop-blur-md text-white text-xs font-extrabold px-4 py-2
rounded-full border border-indigo-400 shadow-xl flex items-center gap-2 pointer-events-auto">
211 |             <Sparkles size={14} className="text-amber-300" />
212 |             '( {highlightedPropertyIds.length} Properties Matched Your Natural Language Query!
213 |         </div>
214 |     )}
215 | </div>
216 | </div>
217 | );
218 | }
219 |

```

10. AI SPATIAL CHATBOT & POI OVERLAY (frontend/src/components/ui/ExplainerChatbot.jsx)

File: frontend/src/components/ui/ExplainerChatbot.jsx | Total Lines: 254

[illegible]

[illegible]

[illegible]


```

179 |         activePoiCategory={poiCategory}}ð
180 |         heightClass="h-[380px]"ð
181 |       />ð
182 |     </div>ð
183 |   ð
184 |   {/ * Chat Section */}ð
185 |   <div className="w-full md:w-[380px] lg:w-[420px] flex flex-col h-[380px] rounded-2xl overflow-hidden
bg-slate-900 text-white shadow-xl">ð
186 |     ð
187 |     {/ * Chat Header */}ð
188 |     <div className="h-14 border-b border-slate-800 flex items-center px-4 bg-slate-950/80">ð
189 |       <Bot size={20} className="text-indigo-400 mr-2" />ð
190 |       <div>ð
191 |         <h3 className="font-bold text-sm text-white">AI Spatial Expert</h3>ð
192 |         <p className="text-[10px] text-indigo-300">Ask & Auto-Pin Malls, Schools, Metro & Hospitals</p>ð
193 |       </div>ð
194 |     </div>ð
195 |   ð
196 |   {/ * Messages */}ð
197 |   <div className="flex-1 overflow-y-auto p-4 space-y-4 relative no-scrollbar">ð
198 |     {messages.map((msg, i) => (ð
199 |       <motion.div ð
200 |         key={i} ð
201 |         initial={{ opacity: 0, y: 5 }} ð
202 |         animate={{ opacity: 1, y: 0 }}ð
203 |         className={`flex ${msg.sender === 'user' ? 'justify-end' : 'justify-start'}`}ð
204 |       >ð
205 |         <div ð
206 |           className={`max-w-[90%] rounded-2xl p-3 text-xs leading-relaxed ${ð
207 |             msg.sender === 'user' ð
208 |               ? 'bg-indigo-600 text-white rounded-br-sm shadow-md font-medium' ð
209 |               : 'bg-slate-800 text-slate-200 rounded-bl-sm border border-slate-700/60'ð
210 |             }`ð
211 |         >ð
212 |           {msg.text.split('\n').map((line, idx) => (ð
213 |             <React.Fragment key={idx}>ð
214 |               {line}ð
215 |               {idx !== msg.text.split('\n').length - 1 && <br />}ð
216 |             </React.Fragment>ð
217 |           )))ð
218 |         </div>ð
219 |       </motion.div>ð
220 |     )))ð
221 |     {loading && (ð
222 |       <div className="flex justify-start">ð
223 |         <div className="bg-slate-800 rounded-2xl rounded-bl-sm p-3 flex gap-1 items-center text-xs
text-indigo-300">ð
224 |           <Loader2 size={14} className="animate-spin text-indigo-400" /> Plotting neighborhood POIs
on map...ð
225 |         </div>ð
226 |       </div>ð
227 |     )}ð
228 |     <div ref={messagesEndRef} />ð
229 |   </div>ð
230 | ð
231 | {/ * Input */}ð
232 | <div className="p-3 bg-slate-950/90 border-t border-slate-800">ð
233 |   <form onSubmit={handleSend} className="relative flex items-center">ð
234 |     <input ð
235 |       type="text" ð
236 |       value={input}ð
237 |       onChange={(e) => setInput(e.target.value)}ð
238 |       placeholder="e.g. Are there malls or schools nearby?"ð
239 |       className="w-full bg-slate-900 border border-slate-700/80 rounded-full py-2.5 pl-4 pr-10
text-xs text-white placeholder:text-slate-400 focus:outline-none focus:border-indigo-500 transition-colors"ð
240 |     />ð
241 |     <button ð
242 |       type="submit" ð
243 |       disabled={!input.trim() || loading}ð
244 |       className="absolute right-2 w-7 h-7 rounded-full bg-indigo-600 flex items-center justify-
center text-white disabled:opacity-50 disabled:cursor-not-allowed hover:bg-indigo-500 transition-colors shadow-md"ð
245 |     >ð
246 |       <Send size={12} />ð
247 |     </button>ð
248 |   </form>ð
249 | </div>ð
250 | ð
251 | </div>ð
252 | </div>ð
253 | );ð
254 | }

```


